Public CAD Is Not Operational Ground Truth

A Hurricane Ida Stress Test of Public Dispatch Data

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Abstract

Researchers commonly measure police response time with timestamps recorded in computer-aided dispatch (CAD) systems. When CAD records are publicly released, that use requires stable field definitions and coverage. In New Orleans, the released record changed abruptly on 28 July 2021. Before that date, none of 419,840 non-officer-initiated records contained an arrival timestamp without a dispatch timestamp. Within two days, that configuration became common, while dispatch-field coverage in officer-initiated records fell from 100 percent to 1 percent. Five weeks later, Hurricane Ida produced a temporary additional shift: among records with a valid arrival field, the public missing-dispatch share reached 43.4 percent on 31 August and 49.3 percent on 1 September. The largest half-day change was 53.2 percentage points, larger than all 151 post-change ordinary-week comparisons. These results document instability in the released measurement product, not physical police response, police performance, or the mechanism generating the public fields. Public CAD timestamps should be used as operational clocks only after field meaning, coverage, and provenance are validated. The validation protocol is directly relevant to environmental-stress studies using CAD call counts, priority groups, or response-time outcomes.

Keywords: calls for service; police responsiveness; administrative data; Hurricane Ida; measurement error; administrative data quality
JEL codes: C18, C55, H12, K42

1 Introduction

Researchers and public agencies commonly construct police response-time measures from timestamps recorded in CAD systems. When CAD records are publicly released, that use assumes the fields retain stable meanings and coverage. This paper tests that assumption in New Orleans before treating the released fields as performance measures. It observes whether public dispatch and arrival fields are present and how their joint pattern changes. A missing dispatch timestamp may reflect a missing operational stage, but it may also reflect workflow, later entry, retention, redaction, or export logic. These field-presence measures describe the released record; they are not response times, physical-event indicators, or measures of service quality.

Two empirical facts organize the paper. First, the New Orleans public file changes abruptly on 28 July 2021. Before that date, no non-officer-initiated record has a valid arrival field without a dispatch field. Beginning on 28 July, that configuration appears and persists. At the same time, dispatch fields almost disappear from officer-initiated records even though their arrival fields remain nearly complete. Second, Hurricane Ida produces an additional, temporary reconfiguration within this new regime. On 31 August and 1 September, the public missing-dispatch share—the share missing a dispatch field among records with a valid arrival field—approaches one-half.

The Ida pattern is large relative to later ordinary weeks. Using every non-officer record in each half-day, the largest event-minus-prior-week change in a public field state is 53.2 percentage points. No post-change ordinary-week comparison is as large. A conditional multinomial bootstrap leaves Ida first in all 4,000 draws, although that exercise does not model time-series dependence. A composition-standardized comparison also ranks Ida first, but it retains incomplete common support and is therefore treated as secondary evidence.

The paper's contribution is about measurement, not about the causal effect of Ida. Public records show that the released measurement product changes before the hurricane and changes again during the event. They do not reveal whether the underlying cause was an operational workaround, an entry convention, a retention rule, or an export transformation. The implication is straightforward: a public CAD timestamp should not be treated automatically as an operational clock. Analysts must first establish the relevant denominator, field meaning, coverage, and lineage.

Environmental shocks can affect both police-service production and the administrative observation process that makes it visible. Observed call volume may change because underlying demand or record-generation rules change. Observed response time may change because service, prioritization, call composition, or endpoint completeness changes. The New Orleans case supplies an empirical reason to validate the observation layer before assigning an operational interpretation.

2 Related literature and contribution

A large economics and policing literature shows that response delays matter. Faster police arrival can improve crime clearance and reduce injuries, while traffic congestion, staffing, and queueing can slow first responders (Blanes i Vidal and Kirchmaier, 2018; DeAngelo et al., 2023; Brent and Beland, 2020; Mourtgos et al., 2024). These studies demonstrate the value of a valid response-time measure. The present paper asks the logically prior question: whether a public administrative export supplies a stable response clock in the first place.

Research on environmental and infrastructure stress asks how congestion, severe weather, disasters, heat, smoke, or pollution alter calls for service, police activity, and first-responder responsiveness. Published studies use administrative records to examine heat-related police activity and 911 medical-dispatch demand (Behrer and Bolotnyy, 2024; Bassil et al., 2008). Federal continuity guidance likewise treats resilient communications and information systems as necessary to sustain essential functions during disruption (Federal Emergency Management Agency, 2024). Substantive studies estimate changes in demand, service production, prioritization, or delay. This paper studies a prerequisite that complements those designs: whether the administrative denominator, response clock, endpoint coverage, call-initiation stream, and priority definition remain stable enough to support the intended construct. Establishing that observation layer strengthens construct validity and helps distinguish operational effects from observation effects (Brent and Beland, 2020; Anastario et al., 2009; Harville et al., 2011).

The paper also relates to work on administrative measurement. Calls-for-service and police records are shaped by reporting, classification, discretion, verification, and data-processing rules (Klinger and Bridges, 1997; Boivin and Cordeau, 2011; Boivin and Ouellet, 2014; Simpson and Orosco, 2021). More generally, measurement error and misclassification can distort group comparisons and structural interpretation (Kapteyn and Ypma, 2007; Chen et al., 2011; Knox et al., 2020). The evidence here adds a time dimension: the joint presence of public CAD fields changes abruptly, so the mapping from activity to released data is not stable across the sample period.

Disaster and domestic-violence research provides substantive motivation. Hurricane exposure, displacement, psychological stress, partner conflict, and demand for services can move together (Anastario et al., 2009; Schumacher et al., 2010; Harville et al., 2011; First et al., 2022). Police composition can affect reporting (Miller and Segal, 2019), and pandemic-era studies show that calls, reports, and underlying victimization need not move one-for-one (Leslie and Wilson, 2020; Bullinger et al., 2021; Miller et al., 2022, 2024). Those studies do not validate a particular CAD timestamp. Ida is used here as a stress test of administrative observability, not as a treatment whose effect on domestic-violence incidence is estimated.

Finally, partial-identification methods provide the appropriate language when public records do not reveal a unique latent process (Manski, 1989, 1990, 2003; Tamer, 2010; Molinari, 2020). The paper does not propose a new general partial-identification estimator. It uses that discipline to distinguish quantities directly observed in the public file from quantities that are selected, bounded, or unavailable without additional evidence.

Relative to these literatures, the paper makes three contributions. It documents a date-localized change in the public CAD measurement regime; it shows that Ida produces an additional extreme but temporary reconfiguration within the later regime; and it translates those facts into a validation protocol for CAD-based studies of public-safety demand, activity, and responsiveness.

3 Data and public field measures

3.1 Official files and analysis population

The analysis uses the official New Orleans Calls for Service files for 2020–2026 (City of New Orleans, 2020, 2021, 2022, 2023, 2024, 2025, 2026). The reproduction package binds the 2020–2024 annual files to official dataset identifiers, row counts, observed date ranges, monthly source hashes, and deterministic annual bundle hashes.

The main analysis is restricted to records coded `SelfInitiated=N`, which the public data distinguish from items generated by officers in the field. These are referred to as *non-officer-initiated* records. Officer-initiated records are analyzed separately because their public dispatch-field convention changes sharply in July 2021.

For each record, define the dispatch field as present when `TimeDispatch` is nonblank. Define the arrival field as valid when `TimeArrive` parses and is no earlier than `TimeCreate`. These two indicators produce four public configurations.

Three descriptive series summarize the public file: the share of records with a dispatch field, the share with a valid arrival field, and the *public missing-dispatch share*—among records with a valid arrival field, the share whose public dispatch field is missing. Calls are grouped by creation date; dispatch and arrival fields are read from the eventual released record.

Table 1. Four configurations in the released public record

Public configuration	Dispatch field	Valid arrival field	What the label does not establish
Neither field	No	No	That no call activity or officer action occurred.
Dispatch only	Yes	No	That no physical arrival occurred.
Arrival only	No	Yes	That physical arrival occurred without dispatch.
Both fields	Yes	Yes	That the endpoint meanings are stable or that the record represents a complete operational pathway.

These labels describe the released record. "Arrival only" does not mean that an officer physically arrived without being dispatched.

4 A public-data regime change on 28 July 2021

The arrival-only configuration is completely absent from non-officer-initiated records before 28 July 2021. Across 419,840 records created from January 2020 through 27 July 2021, its count is exactly zero. It appears abruptly on 28 July and remains present thereafter.

A direct audit of the official July 2021 compressed CSV confirms that the aggregate pipeline did not produce this pattern. All nonblank dispatch and arrival fields in the audit interval parse under the stated rule, and the daily state counts agree exactly with the packaged aggregate tally. Table 2 shows the central transition.

Table 2. The public-file transition, 27–29 July 2021

	27 July	28 July	29 July
Non-officer records	731	620	666
Non-officer arrival-only records	0 (0.0%)	34 (5.5%)	69 (10.4%)
Non-officer dispatch-field share	90.0%	82.1%	76.6%
Officer records	479	575	504
Officer dispatch-field share	100.0%	40.0%	1.0%
Officer valid-arrival share	100.0%	100.0%	100.0%

Percentages are calculated from the official daily source rows. Full state counts for 25–31 July are reported in the empirical supplement.

The simultaneous movement across the two initiation streams is important. Officer-initiated records carry the public initiation label for items generated by officers in the field, yet their public dispatch fields almost disappear while their arrival fields remain complete. The missing public dispatch field therefore cannot be read mechanically as the absence of an officer or of all operational activity. For this stream, the post-break pattern necessarily reflects a change in the mapping from officer-initiated activity to the released public record. The evidence does not identify whether that change occurred in workflow, entry, retention, transformation, or export.

Figure 1 places the break in the longer series. Dispatch-field coverage among non-officer records is about 90 percent in 2020, then declines after July 2021 and reaches 65.4 percent in 2024. The arrival-only configuration, previously absent, becomes increasingly common.

The later annual files confirm that this measurement regime persists beyond the Ida period. In the comparable non-officer denominator, dispatch-field coverage is 65.4 percent in 2024, 66.0 percent in 2025, and 66.8 percent in the 2026 snapshot. The much lower all-row rate of about 41.8 percent combines non-officer records with officer-initiated records, whose dispatch fields are almost always

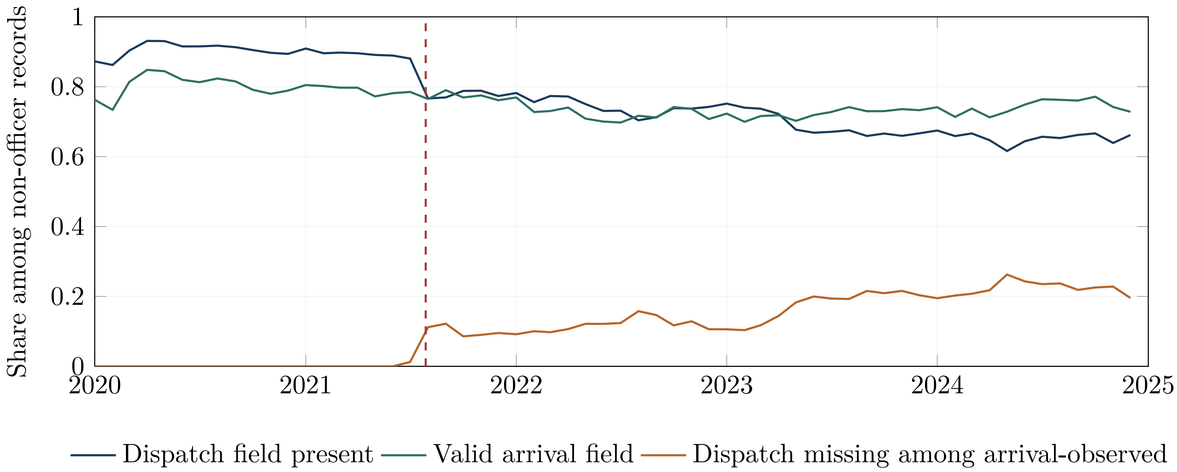

Figure 1. Public-field completeness among non-officer-initiated records. The dashed line marks 28 July 2021, the first date with an arrival-only record.

absent under the later convention.

Table 3. Dispatch-field coverage in the later public-data regime

Year	Non-officer records	Officer records	All public rows
2024	65.4%	1.1%	42.7%
2025	66.0%	0.9%	41.8%
2026 snapshot	66.8%	0.9%	41.9%

The all-row rate is not comparable to the non-officer series because it pools two streams with different recording conventions.

5 Hurricane Ida within the new regime

Hurricane Ida made landfall on 29 August 2021, five weeks after the public-file transition. Figure 2 shows the daily field pattern for non-officer records from one week before Ida through the following two weeks.

During the preceding week, the public missing-dispatch share ranges from 5.1 to 11.7 percent. On landfall day, 29 August, valid-arrival coverage falls to 43.8 percent, compared with 74–84 percent during the preceding week. Dispatch-field coverage remains 72.9 percent. The first event-day change therefore appears mainly in the public arrival field.

The configuration then changes. The public missing-dispatch share rises to 22.9 percent on 30 August, 43.4 percent on 31 August, and 49.3 percent on 1 September as dispatch-field coverage falls to 47.3 and 40.9 percent on the latter two days. It falls to 16.4 percent on 2 September and returns to its preceding range by 5 September, apart from a one-day 10 September excursion to 34.7 percent. The later event days are therefore marked mainly by missing dispatch fields among arrival-observed records.

These movements are large, but their interpretation must remain administrative. The figures index calls by creation date and read fields from the eventual released record. They do not show a live system snapshot, nor do they establish that a unit physically arrived without dispatch. They show that records created during the event were ultimately released in a different field configuration.

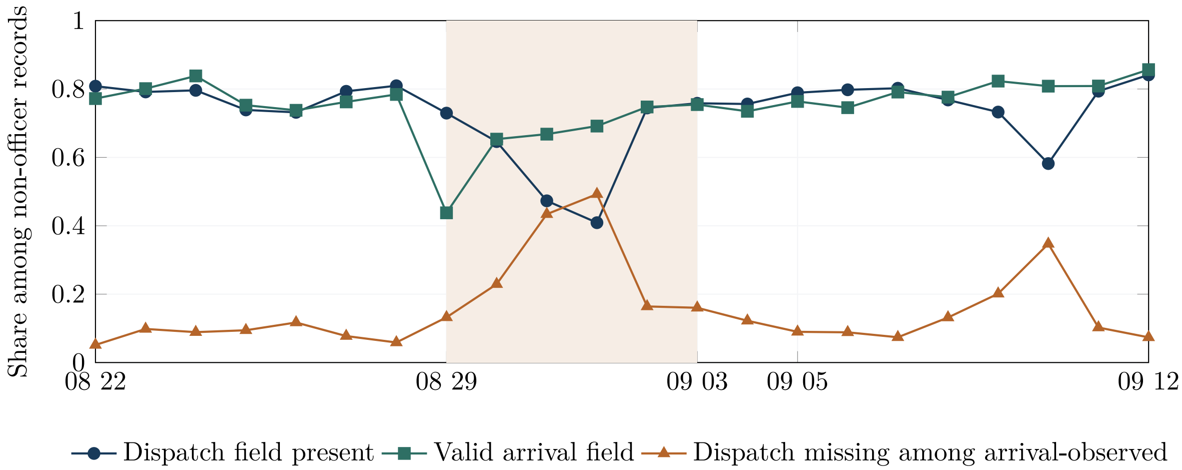

Figure 2. The public-field path around Hurricane Ida. Shading marks the five-day event window. The figure describes the released record, not physical dispatch or arrival.

6 How unusual was Ida?

6.1 Primary full-count comparison

For each five-day window, the analysis compares ten twelve-hour cells with the same half-days one week earlier. Within each cell it computes changes in three of the four public configurations: dispatch only, arrival only, and both fields present. The neither-field share is determined by the other three because the four shares sum to one. The summary statistic is the largest absolute change across the thirty cell-by-state comparisons:

$$M(w) = \max_{b,s} \left| \widehat{p}_{w,b,s} - \widehat{p}_{w-7,b,s} \right| . \tag{1}$$

For Ida, the largest public-state share changed by 53.2 percentage points ($M = 0.532$).

The primary reference comparison uses 151 ordinary windows for which both the event and baseline periods occur after 28 July 2021. Prespecified emergency and holiday windows remain excluded. This post-change set was defined after the July transition was identified, so it is a transparent descriptive sensitivity rather than a preregistered test.

The largest reference value is 0.310. Thus Ida is larger than all 151 post-change ordinary-week comparisons. A conditional window-wise multinomial bootstrap independently perturbs Ida and the reference windows. Ida ranks first in all 4,000 draws, with a 95-percent interval of $[0.475, 0.601]$ for its statistic. Because the procedure does not preserve serial or cross-window dependence, it is best read as a robustness check against ordinary categorical-count variation, not as formal time-series inference.

6.2 Secondary standardized comparison

A second estimator standardizes across common call-type, hour, and schema strata. It also ranks Ida first: the maximum standardized contrast is 0.507, compared with 0.328 for the largest of 153 prespecified references beginning on or after 1 July 2021. The supplement calls this the stage-era set and documents its inherited cutoff. However, the standardized analysis covers only 86.1 percent of event records and 77.0 percent of baseline records. Ida would fail the 0.90 support rule applied to candidate reference windows. The standardized result is therefore secondary; the full-count comparison is the main evidence.

Table 4. Ida relative to ordinary-week comparisons

Design	References	Ida largest change	Largest reference	Interpretation
Full-count, post-change	151	53.2 pp	31.0 pp	Primary descriptive comparison; uses every non-officer record in each half-day.
Standardized, stage-era	153	50.7 pp	32.8 pp	Same qualitative rank, but incomplete common support makes it secondary.

Two excluded episodes help interpret the result. The week after Ida has a large statistic of 0.478 because its baseline is Ida itself; it captures recovery rather than an independent shock. Hurricane Francine, a later-regime hurricane in September 2024, has a much smaller statistic of 0.110. These later-regime comparisons separate the Ida reconfiguration from a generic hurricane-week contrast.

All other prespecified statistics and threshold variants are reported in the empirical supplement. They do not all rank Ida first. The main conclusion is narrow: Ida produces an unusually large change in the public field configuration.

7 What do the data suggest—and what remains unknown?

7.1 A recording-convention clue

The officer-initiated stream provides the most concrete clue. The public definition describes these items as generated by officers in the field. Before 28 July 2021, their dispatch and arrival fields are almost universally present. After the break, arrival remains nearly universal while dispatch falls to approximately 1 percent. The arrival-only configuration is therefore a normal released form for records carrying that public initiation label under the later regime.

During Ida, non-officer records move toward the same public configuration. One plausible pathway is that some calls normally represented with a dispatch field were entered, retained, or exported through a workflow that omitted it. That possibility is consistent with a change in how activity is represented in the released record, but it is not identified. Internal call-origin provenance, event-entry histories, unit-assignment histories, and export logs would be required to distinguish reclassification, operational workarounds, back-filling, and export omission.

7.2 Disposition mix is not the main accounting explanation

The late Ida movement also occurs within recorded final-disposition categories. On the afternoon and evening of 31 August, the public dispatch-field share among arrival-observed records falls by 49.5 percentage points. A change in the observed disposition mix would have increased the share by about 1.2 points; within-disposition changes account for approximately −50.7 points. The following morning produces nearly the same decomposition. The observed final-disposition mix therefore does not explain the aggregate decline.

This is an accounting result, not a mechanism test. The analysis conditions on records with arrival fields and uses final recorded dispositions. Selection into that population, changes within broad categories, and changes in recording practice remain possible.

7.3 What public documents establish

Official metadata define the public field labels and identify the data provider. The DataNOLA page also warns that records are preliminary, may contain mechanical or human error, may change after investigation, and should not be used mechanically for comparison over time (City of New Orleans, 2021). Public oversight materials document the Ida emergency context (Office of the Independent Police Monitor, 2021), while NOPD policy materials describe intended communications and records practices (New Orleans Police Department, 2023).

The public chronology rules out two proposed technology explanations. The publicly indexed Carbyne APEX cutover occurred in June 2022, eleven months after the July 2021 change (Orleans Parish Communication District, 2022). The Hexagon OnCall Records project was contracted but never launched and therefore cannot explain the transition (New Orleans Office of Inspector General, 2024). These exclusions are useful, but they do not identify the actual cause.

No public versioned changelog or internal-to-public reconciliation was available for this analysis, and no agency confirmation of the July change was obtained. A focused public-records request to OPCD for the CAD export configuration and change history covering July 2021 is the most direct next step for resolving the mechanism. Until such evidence is available, the strongest supported conclusion is a discontinuity in the released record, not a named institutional mechanism.

8 Validating police-service measures under environmental and system stress

The paper's measurement lesson applies beyond Hurricane Ida and provides a low-cost validation protocol for CAD-based research under environmental and system stress. Shocks can affect both the production of police service and the observation process. To support interpretation, an analysis should declare the denominator and call-initiation rule, define the response clock, report endpoint coverage, and document the priority definition and data lineage.

Table 5. Low-cost validation for downstream outcomes

Downstream outcome	Measurement question	Low-cost validation	What the validation supports
Call volume	Are documented initiation streams separated and stable?	Report the initiation rule, duplicate and cancellation treatment, and counts by documented stream.	Supports interpretation as demand rather than record production.
Response time	What are the start and end points of the response clock, and what share of calls contains both?	Report endpoint coverage by exposure group, priority, and call type.	Establishes the complete-case population and construct validity.
Priority split	Is priority initial, final, or updated during processing?	Report priority shares and, where available, initial-to-final transitions.	Separates service prioritization from changing classification.
Police activity/coverage	Are fields and extraction rules stable over time?	Audit schema continuity and known system changes.	Strengthens longitudinal comparability and reproducibility.

The New Orleans evidence directly motivates the first two checks: initiation streams follow different public-field conventions, and endpoint coverage changes sharply over time and during Ida. The priority and police-activity checks are adjacent extensions for studies that use those outcomes.

These checks complement substantive estimates; they do not turn endpoint presence alone into a measure of queueing or capacity. Reported DV is one high-stakes downstream application. Reported CAD calls are not equivalent to underlying domestic-violence incidence or complete help-seeking. A future DV analysis should preserve unresolved classifications rather than force every record into or out of the target group, and it should report endpoint coverage before interpreting observed durations. The supplement reports the corresponding classification and missing-data bounds. Ida motivates this measurement work, but the current system-level analysis does not estimate a DV effect.

9 Limitations

The analysis uses eventual public records rather than internal event histories. The July change is verified in the official source bytes, but its institutional cause is not observed. The post-change comparison set was defined after the break was identified and excludes specified emergency and holiday contexts. Reference windows overlap; retaining every second window removes adjacent overlap and leaves Ida first in both alternating phases (supplement). The bootstrap does not preserve all time-series dependence. Public `SelfInitiated=N` is not equivalent to "911 only." Weather, holidays, mobility, evacuation, reporting behavior, call composition, suppression, and other contemporaneous conditions can affect the observed record.

These limitations restrict interpretation but do not erase the descriptive result. The released field configuration changes abruptly in July 2021, changes further during Ida, and follows different conventions across initiation streams.

10 Conclusion

Public CAD files are valuable administrative records, but their timestamps are not operational ground truth by definition. In New Orleans, a new arrival-only configuration appears suddenly on 28 July 2021, accompanied by an almost complete disappearance of dispatch fields from officer-initiated records. Five weeks later, Hurricane Ida produces an additional two-day reconfiguration that is larger than every post-change ordinary-week comparison.

The public evidence identifies the timing and magnitude of changes in the released record. It does not identify the underlying operational sequence or the institutional mechanism that produced those fields. Analysts should therefore treat public CAD timestamps as administrative measures until their meanings, coverage, and lineage have been validated for the specific performance question.

The same validation logic applies when heat, smoke, pollution, outages, or disasters are used to study police-service demand and responsiveness.

Data and code availability

Replication materials, source bindings, the empirical supplement, and the release verifier are available in the public repository at https://github.com/yuwen3756-hue/public-cad-measurement-police-responsiveness-hurricane-ida. The repository provides derived aggregate files and reproducible code; official raw CAD files remain identified by first-hand public-source links and hashes in the reproduction materials.

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